Molecular dynamics introduction

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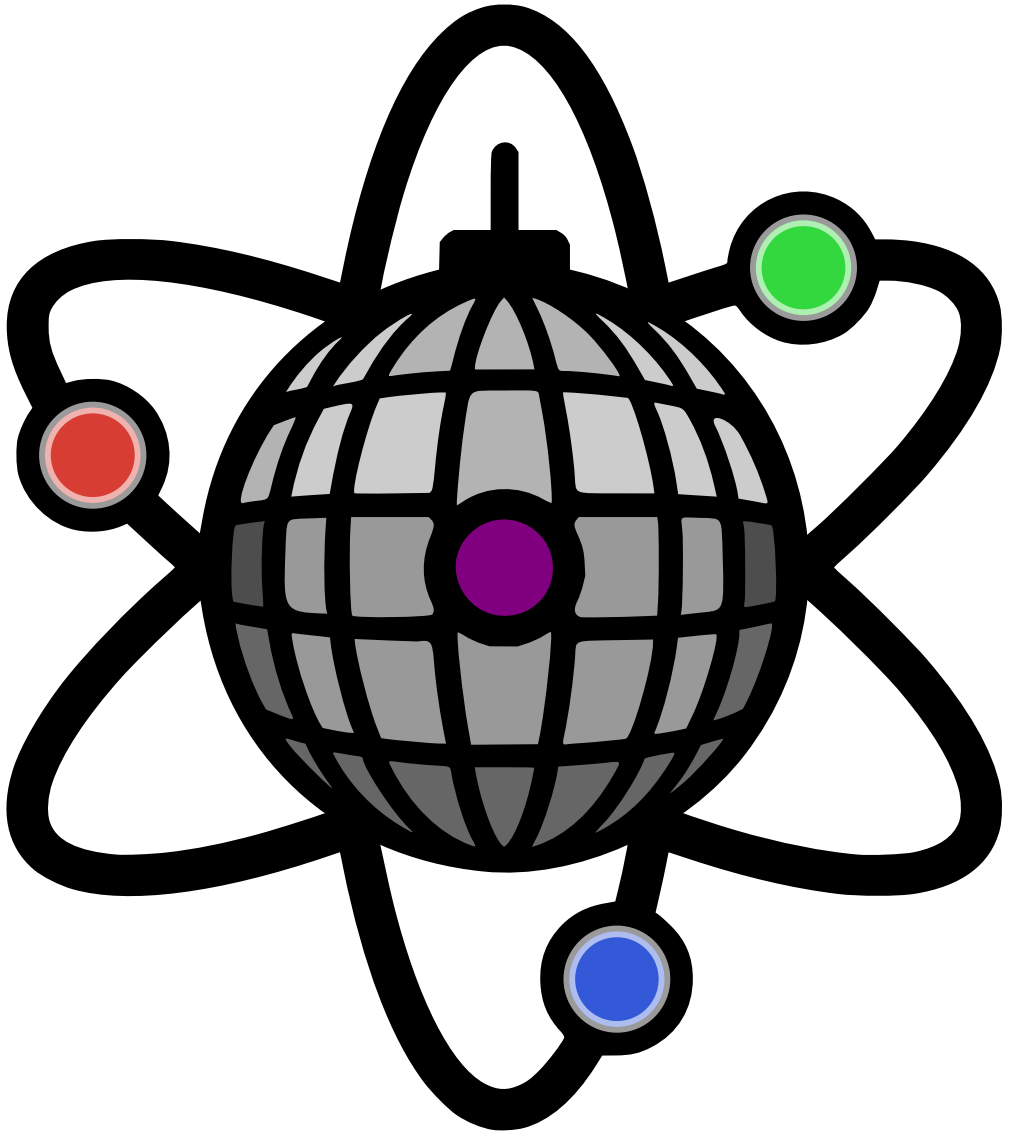
MDANSE Workshop 2026

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Molecular dynamics introduction

- Statistical mechanics
- Monte-Carlo
- Equations of motion
- Ergodicity (NVE)
- Temperature
- Questions



Statistical mechanics

- The aim of molecular dynamics is to solve numerically the ensemble averages for specific properties. For example, with the microcanonical ensemble (fixed number of atom, volume, and energy).

Average property of a

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

Statistical mechanics

- The aim of molecular dynamics is to solve numerically the ensemble averages for specific properties. For example, with the microcanonical ensemble (fixed number of atom, volume, and energy).

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

Partition function of the microcanonical ensemble

Statistical mechanics

- The aim of molecular dynamics is to solve numerically the ensemble averages for specific properties. For example, with the microcanonical ensemble (fixed number of atom, volume, and energy).

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

Hamiltonian

Statistical mechanics

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$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$



Phase space coordinates (positions and momenta)

Statistical mechanics

- The aim of molecular dynamics is to solve numerically the ensemble averages for specific properties. For example, with the microcanonical ensemble (fixed number of atom, volume, and energy).

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$



Energy

Statistical mechanics

- The aim of molecular dynamics is to solve numerically the ensemble averages for specific properties. For example, with the microcanonical ensemble (fixed number of atom, volume, and energy).

Value of property a at phase space coordinate x

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

Statistical mechanics

- The aim of molecular dynamics is to solve numerically the ensemble averages for specific properties. For example, with the microcanonical ensemble (fixed number of atom, volume, and energy).



$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

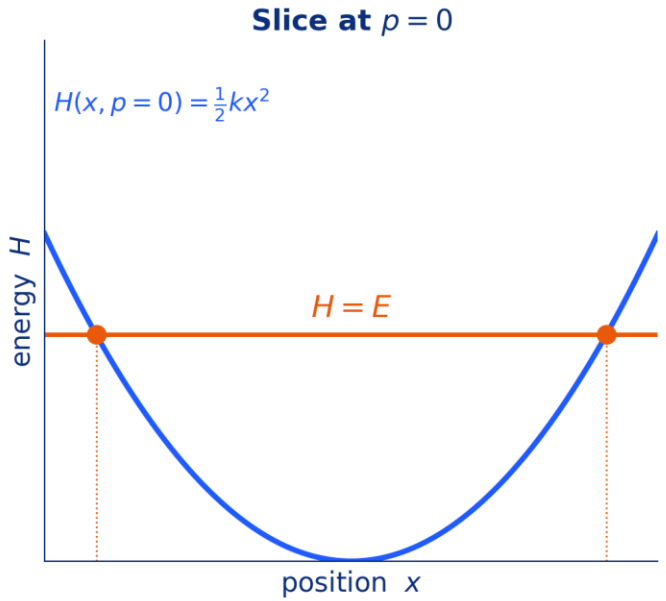
$$\int \mathrm{d}\mathbf{x}\, a(\mathbf{x})\, \delta\big(H(\mathbf{x}) - E\big) = \oint_{H(\mathbf{x})=E} \frac{\mathrm{d}S}{|\nabla H(\mathbf{x})|} a(\mathbf{x})$$

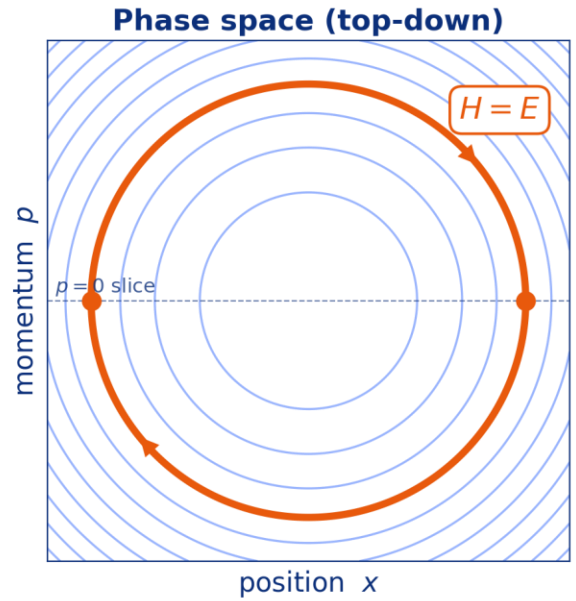
Statistical mechanics

- So, in the microcanonical ensemble the ensemble average for a specific property can be determined via the integral over the phase space coordinates or equivalently an integral over the constant energy phase space surface.
- For the harmonic oscillator, this is a line integral over the constant energy path.

NVE

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$



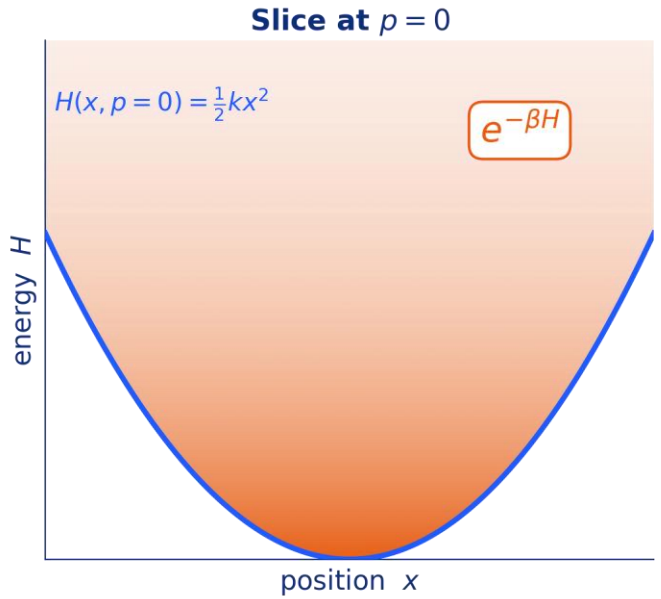


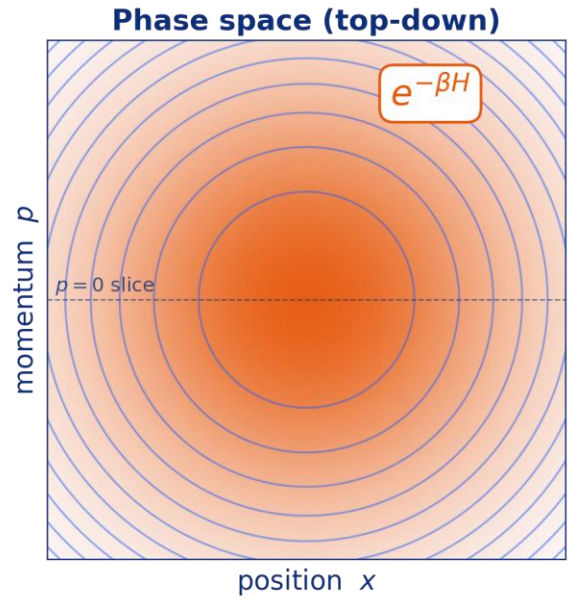
Statistical mechanics

- For canonical ensemble, the ensemble average is a weighted average over the phase space coordinates.

NVT

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x}) e^{-\beta H(\mathbf{x})}}{\int \mathrm{d}\mathbf{x}\ e^{-\beta H(\mathbf{x})}}$$





Statistical mechanics

- Remember that in the thermodynamic limit (large number of particles and large volume) all ensemble averages are equivalent, and we can choose the ensemble which is most convenient.
- However, since we can never simulate a system at the thermodynamic limit there will be difference between the average with the different ensembles.

NVE

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

NVT

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})e^{-\beta H(\mathbf{x})}}{\int \mathrm{d}\mathbf{x}\ e^{-\beta H(\mathbf{x})}}$$

NPT

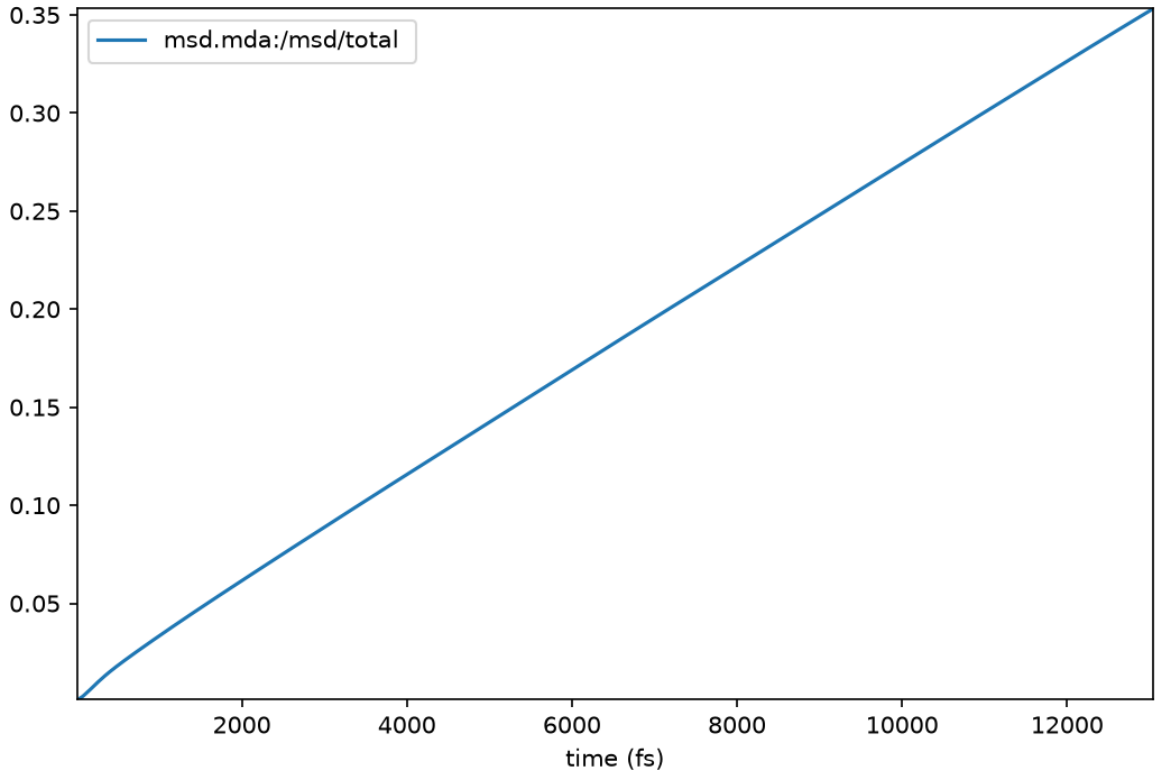
$$\langle a \rangle = \frac{\int \mathrm{d}V\mathrm{d}\mathbf{x}\ a(\mathbf{x}, V)e^{-\beta[H(\mathbf{x},V)+PV]}}{\int \mathrm{d}V\mathrm{d}\mathbf{x}\ e^{-\beta[H(\mathbf{x},V)+PV]}}$$

Statistical mechanics

- Generally, the type of the ensemble should be chosen to reflect an experiment or chosen based on the property we want to calculate.
- For example, with MD we may want to use NVE for dynamics calculations in MD as the thermostats in NVT or NPT will interfere with the trajectory velocities.

NVE

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$



Monte-Carlo

- To understand how we numerically integrate the ensemble averages in statistical mechanics with MD. Let's quickly see how it's done in Monte-Carlo methods.



Salle Europe, Monte Carlo Casino / wikipedia

Monte-Carlo

- First let's say that we property b(**r**) which we are interested in which only depends on positions so that we can integrate away the coordinates over the momenta.

NVT $$\langle b \rangle = \frac{\int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) e^{-\beta U(\mathbf{r})}}{\int \mathrm{d}\mathbf{r}\ e^{-\beta U(\mathbf{r})}}$$

Monte-Carlo

- First let's say that we property b(**r**) which we are interested in which only depends on positions so that we can integrate away the coordinates over the momenta.
- We can replace the ensemble average with a summation. Where the sampling of **r**_i must be drawn from a distribution following f(**r**).

NVT $$\langle b \rangle = \frac{\int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) e^{-\beta U(\mathbf{r})}}{\int \mathrm{d}\mathbf{r}\ e^{-\beta U(\mathbf{r})}}$$

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) f(\mathbf{r}) = \lim_{M \to \infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

$$f(\mathbf{r}) = \frac{e^{-\beta U(\mathbf{r})}}{\int \mathrm{d}\mathbf{r}\ e^{-\beta U(\mathbf{r})}}$$

Central limit theorem

- Why can we replace this integral with a summation?

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) f(\mathbf{r}) = \lim_{M \to \infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

Central limit theorem

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) f(\mathbf{r}) = \lim_{M \to \infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

- Why can we replace this integral with a summation?
- Say we sample a distribution function M times to obtain an approximation of the average value of property some property.

$$\frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i) = b'_0$$

Central limit theorem

- Why can we replace this integral with a summation?
- Say we sample a distribution function M times to obtain an approximation of the average value of property some property.
- If we do this again multiple times we obtain different values for our average.

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) f(\mathbf{r}) = \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

$$\frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i) = b'_0$$

$$\frac{1}{M} \sum_{j=1}^{M} b(\mathbf{r}_j) = b'_1$$

$$\frac{1}{M} \sum_{k=1}^{M} b(\mathbf{r}_k) = b'_2$$

$$\vdots$$

Central limit theorem

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\; b(\mathbf{r}) f(\mathbf{r}) = \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

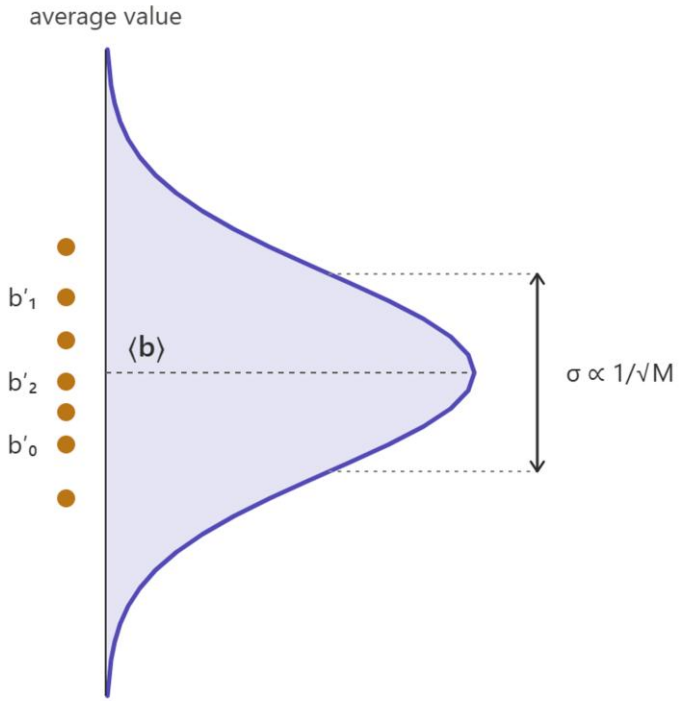
- Why can we replace this integral with a summation?
- Say we sample a distribution function M times to obtain an approximation of the average value of property some property.
- If we do this again multiple times we obtain different values for our average.
- The distribution of these average will follow a Gaussian normal distribution around <b> with a standard deviation which goes as 1 / M^(1/2).

$$\frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i) = b'_0$$

$$\frac{1}{M} \sum_{j=1}^{M} b(\mathbf{r}_j) = b'_1$$

$$\frac{1}{M} \sum_{k=1}^{M} b(\mathbf{r}_k) = b'_2$$

$$\vdots$$



Monte-Carlo

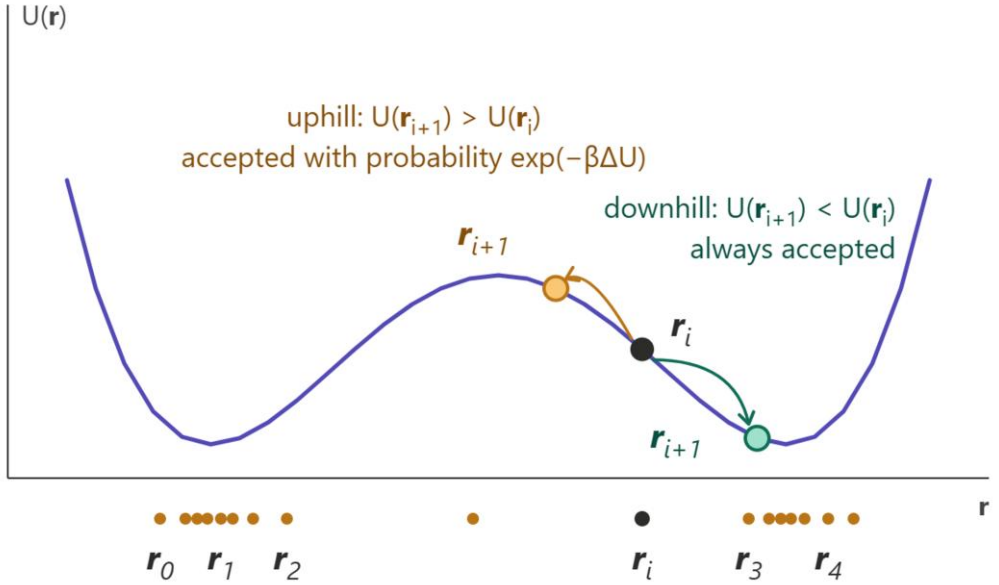
- Algorithms such as Metropolis-Hastings can be used to generate a chain of **r**_i samples.
- This works by generating a trail sample **r**' and then sampling the trail **r**' or resampling the current **r** based on a specific acceptance criteria.

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) f(\mathbf{r}) = \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

$$f(\mathbf{r}) = \frac{e^{-\beta U(\mathbf{r})}}{\int \mathrm{d}\mathbf{r}\ e^{-\beta U(\mathbf{r})}}$$

$$A(\mathbf{r}'|\mathbf{r}) = \min\left[1, e^{-\beta[U(\mathbf{r}')-U(\mathbf{r})]}\right]$$

- With this acceptance criteria, at the limit of an infinite chain of samples, the samples will be distributed following f(**r**).



Monte-Carlo

- As you can see in the Monte-Carlo section, we are simply solving the ensemble averages numerically.
- Mostly importantly it tries to samples **r** so that they are distributed following f(**r**).

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) f(\mathbf{r}) = \lim_{M \to \infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

$$f(\mathbf{r}) = \frac{e^{-\beta U(\mathbf{r})}}{\int \mathrm{d}\mathbf{r}\ e^{-\beta U(\mathbf{r})}}$$

Monte-Carlo

- As you can see in the Monte-Carlo section, we are simply solving the ensemble averages numerically.
- Mostly importantly it tries to samples **r** so that they are distributed following f(**r**).
- In molecular dynamics, we generate a trajectory by evolving the positions and velocities over time by solving the equations of motion numerically.
- However, it is not immediately clear how this can be used to calculate an ensemble average, especially so for the NVT and NPT ensembles.

NVE

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

NVT

$$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})e^{-\beta H(\mathbf{x})}}{\int \mathrm{d}\mathbf{x}\ e^{-\beta H(\mathbf{x})}}$$

NPT

$$\langle a \rangle = \frac{\int \mathrm{d}V\mathrm{d}\mathbf{x}\ a(\mathbf{x}, V)e^{-\beta[H(\mathbf{x},V)+PV]}}{\int \mathrm{d}V\mathrm{d}\mathbf{x}\ e^{-\beta[H(\mathbf{x},V)+PV]}}$$

Equations of motion

- In molecular dynamics we numerically solve the equations of motions and generate a discretized trajectory of atom positions and velocities.
- There are many iterative methods which can be used to solve the ordinary differential equations on the right. We will go through the Euler and Verlet methods.

$$\frac{\mathrm{d}q(t)}{\mathrm{d}t} = \frac{1}{m}p(t)$$

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = -\frac{\partial U}{\partial q(t)}$$

Equations of motion

- In the Euler method we approximate the left side of both differential equations with a first order forward difference.

Euler method

$$\frac{\mathrm{d}q(t)}{\mathrm{d}t} = \frac{1}{m}p(t) \quad \Rightarrow \quad \frac{q(t+\Delta t) - q(t)}{\Delta t} = \frac{1}{m}p(t)$$

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = -\frac{\partial U}{\partial q(t)} \quad \Rightarrow \quad \frac{p(t+\Delta t) - p(t)}{\Delta t} = ma(t)$$

Equations of motion

- Some algebra gives us two sets of equations which allows us to iterate over the equations of motions to obtain a time series of atom positions and velocities.
- In the Euler method both the positions and velocities are updated at the same time.

Euler method

$$q(t + \Delta t) = q(t) + v(t)\Delta t$$

$$v(t + \Delta t) = v(t) + a(t)\Delta t$$

Equations of motion

Verlet integration

- Alternatively, if we rewrite the equations of motion as a single second-order ODE.

$$\frac{\mathrm{d}^2 q(t)}{\mathrm{d}t^2} = a(t)$$

Equations of motion

- Alternatively, if we rewrite the equations of motion as a single second-order ODE.
- In the Verlet integration we approximate the left side with second-order central finite difference.

Verlet integration

$$\frac{\mathrm{d}^2 q(t)}{\mathrm{d}t^2} = a(t)$$

$$\frac{q(t+\Delta t) - 2q(t) + q(t-\Delta t)}{\Delta t^2} = a(t)$$

Equations of motion

- Alternatively, if we rewrite the equations of motion as a single second-order ODE.
- In the Verlet integration we approximate the left side with second-order central finite difference.
- If we do a bit of algebra, we can get the Verlet integration for position updates.

Verlet integration

$$q(t + \Delta t) = 2q(t) - q(t - \Delta t) + a(t)\Delta t^2$$

Equations of motion

- Alternatively, if we rewrite the equations of motion as a single second-order ODE.
- In the Verlet integration we approximate the left side with second-order central finite difference.
- If we do a bit of algebra, we can get the Verlet integration for position updates.
- We can get velocities via a central difference.

Verlet integration

$$q(t + \Delta t) = 2q(t) - q(t - \Delta t) + a(t)\Delta t^2$$

$$v(t) = \frac{q(t + \Delta t) - q(t - \Delta t)}{2\Delta t}$$

Equations of motion

- Alternatively, if we rewrite the equations of motion as a single second-order ODE.
- In the Verlet integration we approximate the left side with second-order central finite difference.
- If we do a bit of algebra, we can get the Verlet integration for position updates.
- We can get velocities via a central difference. With some more algebra to get the velocity-Verlet.

Velocity-Verlet integration

$$q(t + \Delta t) = q(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2$$

$$v(t + \Delta t) = v(t) + \frac{\Delta t}{2}[a(t) + a(t + \Delta t)]$$

Equations of motion

- Although slightly more complicated, the Verlet integrator is the preferred out of the two. The Verlet integrator is time-reversable while Euler is not.

Euler method

	Forwards	Backwards
$T = t - \Delta t$	$q(t) = q(t - \Delta t) + v(t - \Delta t)\Delta t$	$q(t - 2\Delta t) = q(t - \Delta t) - v(t - \Delta t)\Delta t$
$T = t$	$q(t + \Delta t) = q(t) + v(t)\Delta t$	$q(t - \Delta t) = q(t) - v(t)\Delta t$
$T = t + \Delta t$	$q(t + 2\Delta t) = q(t + \Delta t) + v(t + \Delta t)\Delta t$	$q(t) = q(t + \Delta t) - v(t + \Delta t)\Delta t$

Equations of motion

- Although slightly more complicated, the Verlet integrator is the preferred out of the two. The Verlet integrator is time-reversable while Euler is not.

Euler method

Forwards

$$q(t + \Delta t) = q(t) + v(t)\Delta t$$

Backwards

$$q(t + \Delta t) = q(t) + v(t + \Delta t)\Delta t$$

Equations of motion

- Although slightly more complicated, the Verlet integrator is the preferred out of the two. The Verlet integrator is time-reversable while Euler is not.

Verlet integration

	Forwards	Backwards
$T = t - \Delta t$	$q(t) = 2q(t-\Delta t) - q(t-2\Delta t) + a(t-\Delta t)\Delta t^2$	$q(t-2\Delta t) = 2q(t-\Delta t) - q(t) + a(t-\Delta t)\Delta t^2$
$T = t$	$q(t+\Delta t) = 2q(t) - q(t-\Delta t) + a(t)\Delta t^2$	$q(t-\Delta t) = 2q(t) - q(t+\Delta t) + a(t)\Delta t^2$
$T = t + \Delta t$	$q(t+2\Delta t) = 2q(t+\Delta t) - q(t) + a(t+\Delta t)\Delta t^2$	$q(t) = 2q(t+\Delta t) - q(t+2\Delta t) + a(t+\Delta t)\Delta t^2$

Equations of motion

- Although slightly more complicated, the Verlet integrator is the preferred out of the two. The Verlet integrator is time-reversable while Euler is not.

Verlet integration

Forwards

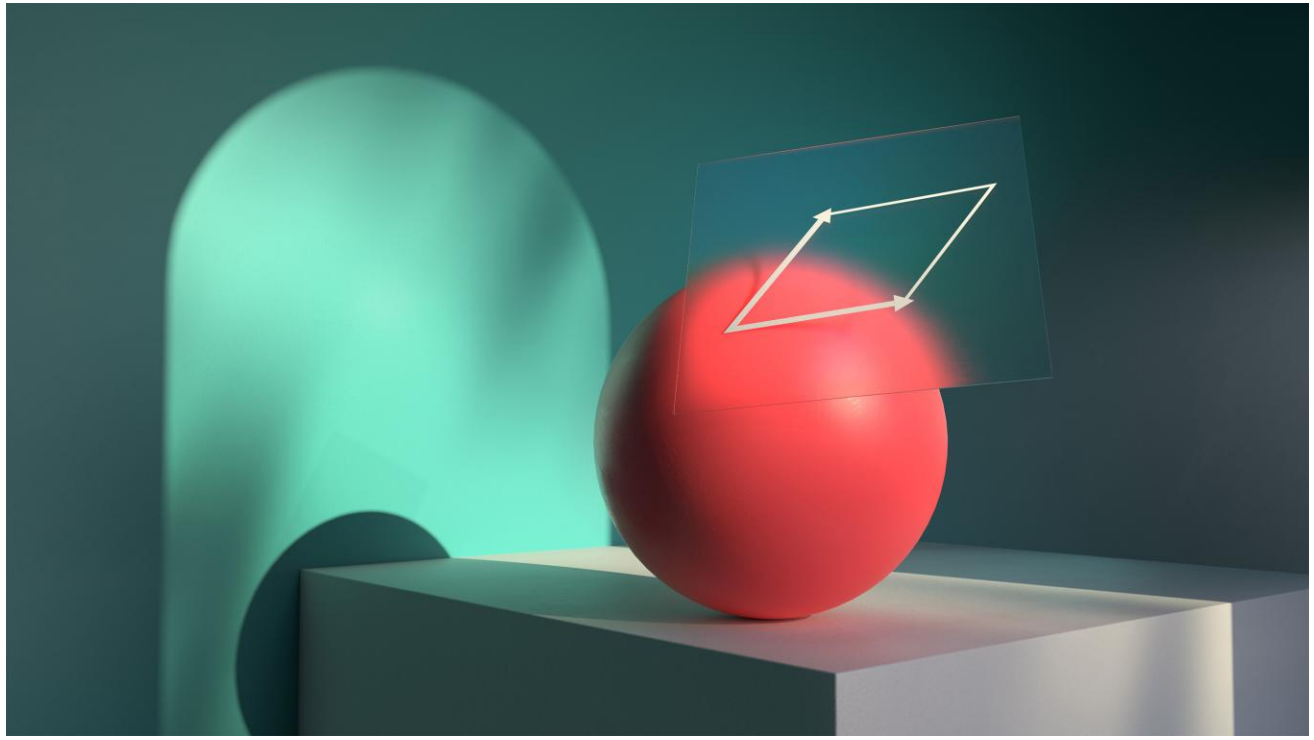
$$q(t + \Delta t) = 2q(t) - q(t - \Delta t) + a(t)\Delta t^2$$

Backwards

$$q(t + \Delta t) = 2q(t) - q(t - \Delta t) + a(t)\Delta t^2$$

Equations of motion

- The Verlet integrator is also a symplectic integrator and preserves the symplectic properties of classical mechanics.
- But wait what is symplectic? To answer this, we'd need to introduce some abstract mathematics beyond me and this introductory class on MD.
- The symplectic property of classical mechanics defines how the phase-space flows over time and is a stricter property than the incompressibility of phase-space.



Olena Shmahalo/Quanta Magazine

Equations of motion

- The Verlet integrator therefore preserves phase-space volume as a consequence of it being a symplectic integrator.
- The Euler method is neither a symplectic integrator nor does it preserve the phase-space volume.
- This can be seen even with a simulation using the harmonic oscillator.

Euler method

$$q(t+\Delta t) = q(t) + v(t)\Delta t$$

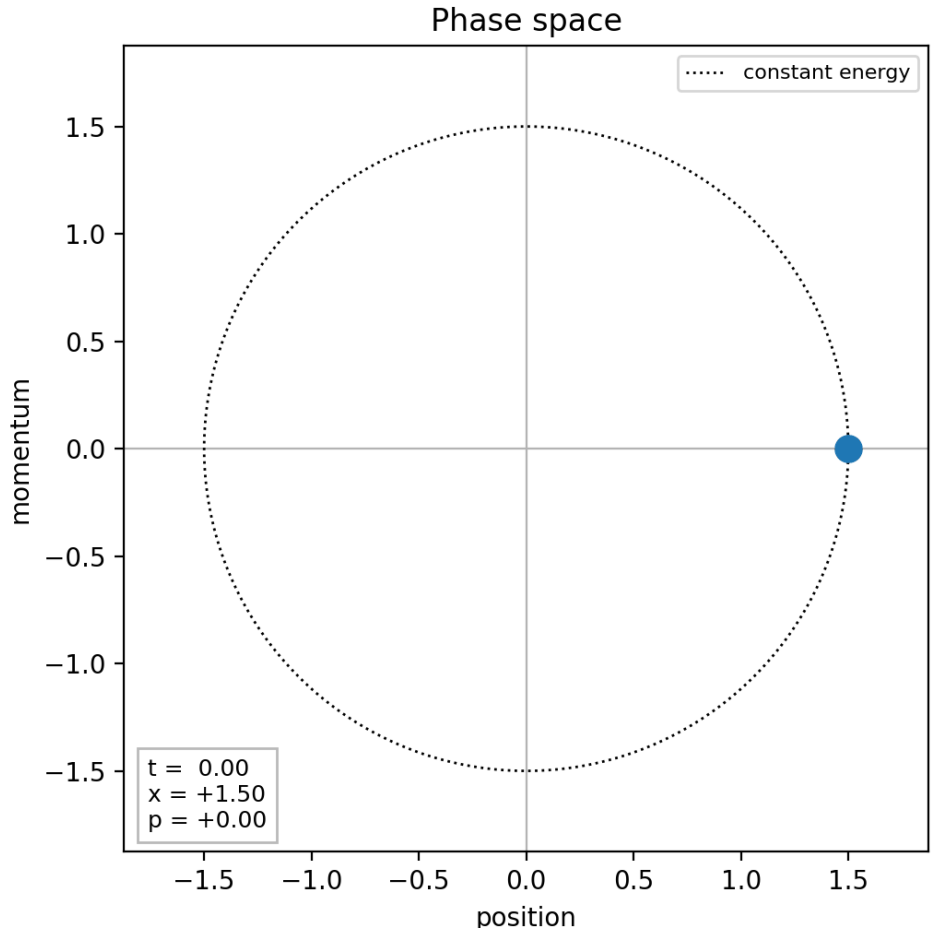
$$v(t+\Delta t) = v(t) + a(t)\Delta t$$

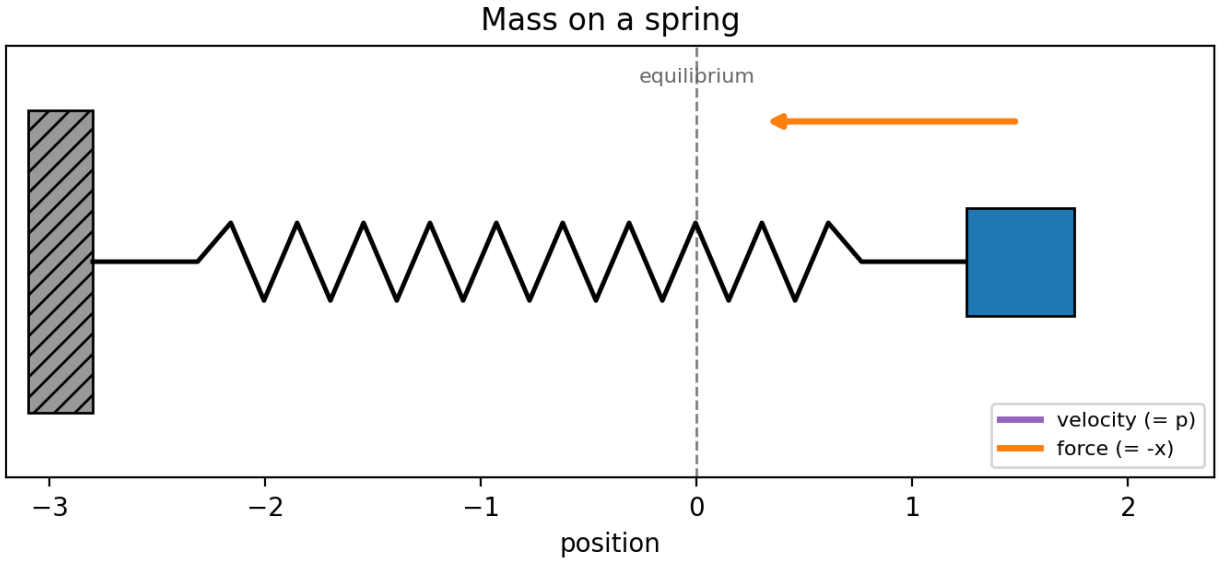
Velocity-Verlet

$$q(t+\Delta t) = q(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2$$

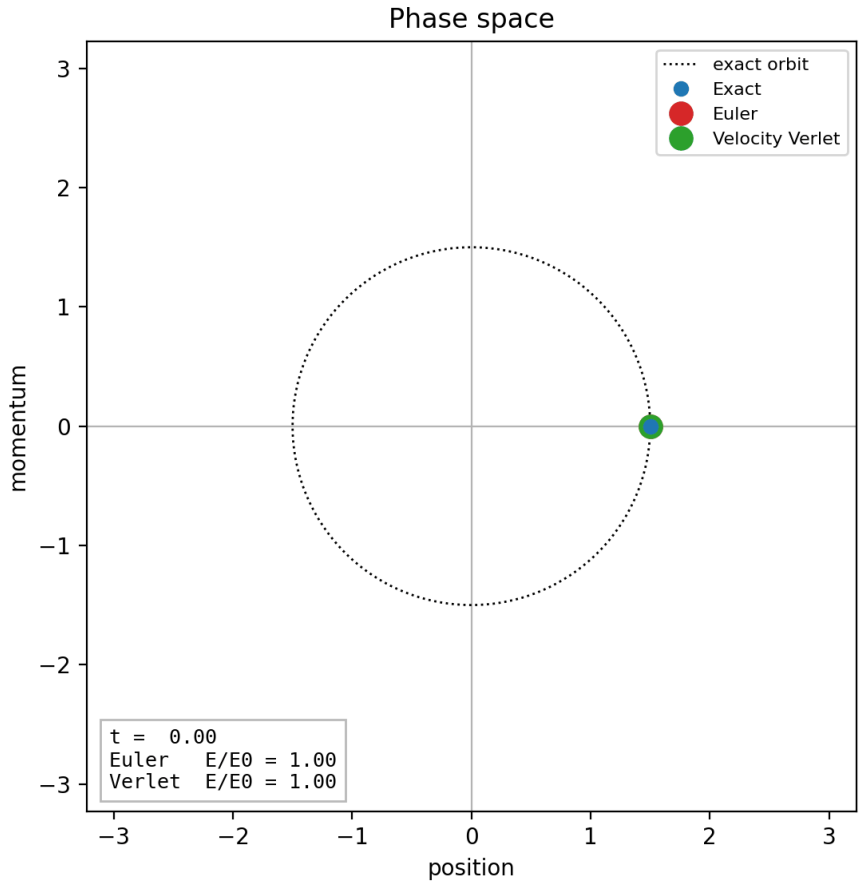
$$v(t+\Delta t) = v(t) + \frac{\Delta t}{2}[a(t) + a(t+\Delta t)]$$

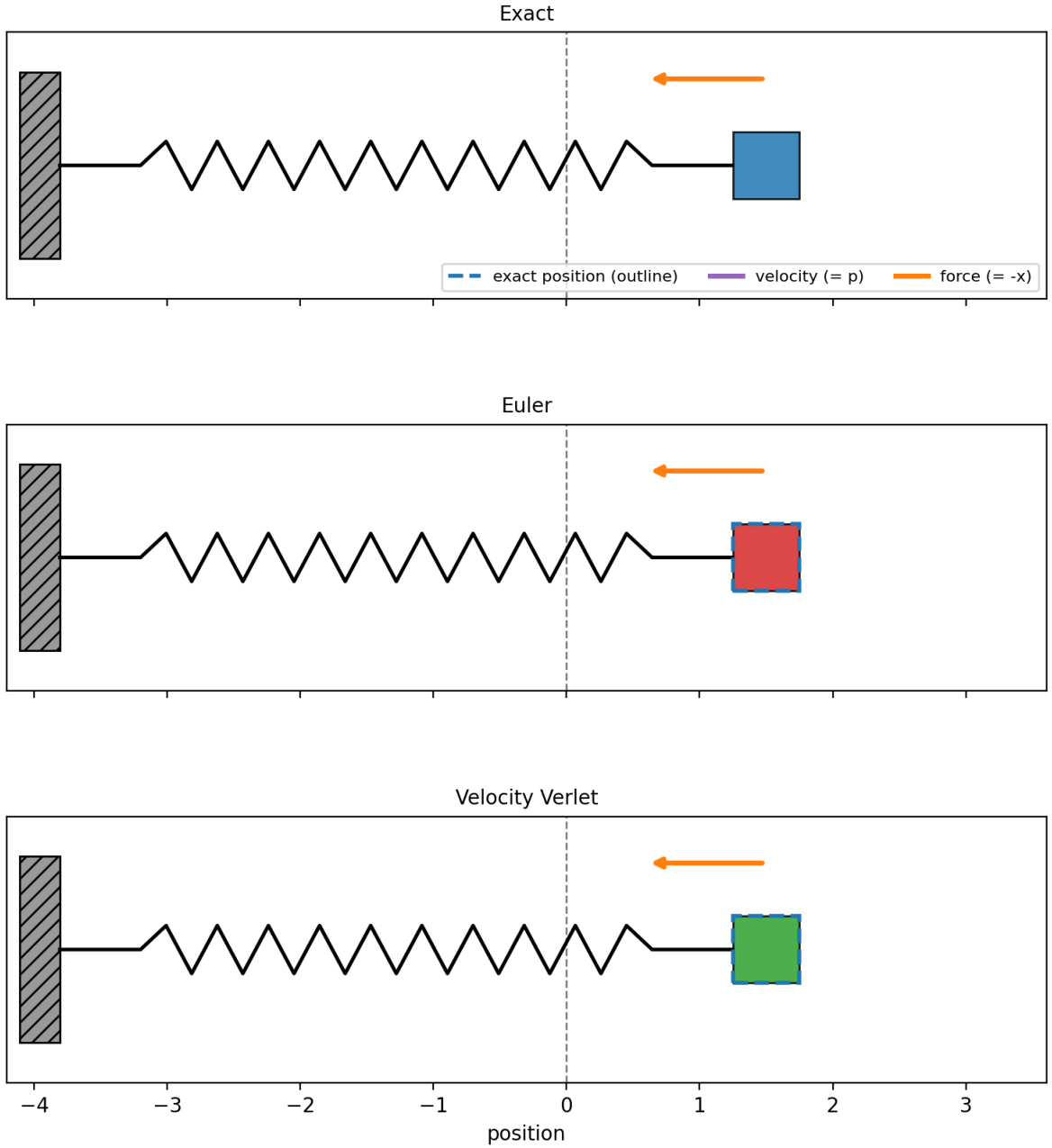
Equations of motion



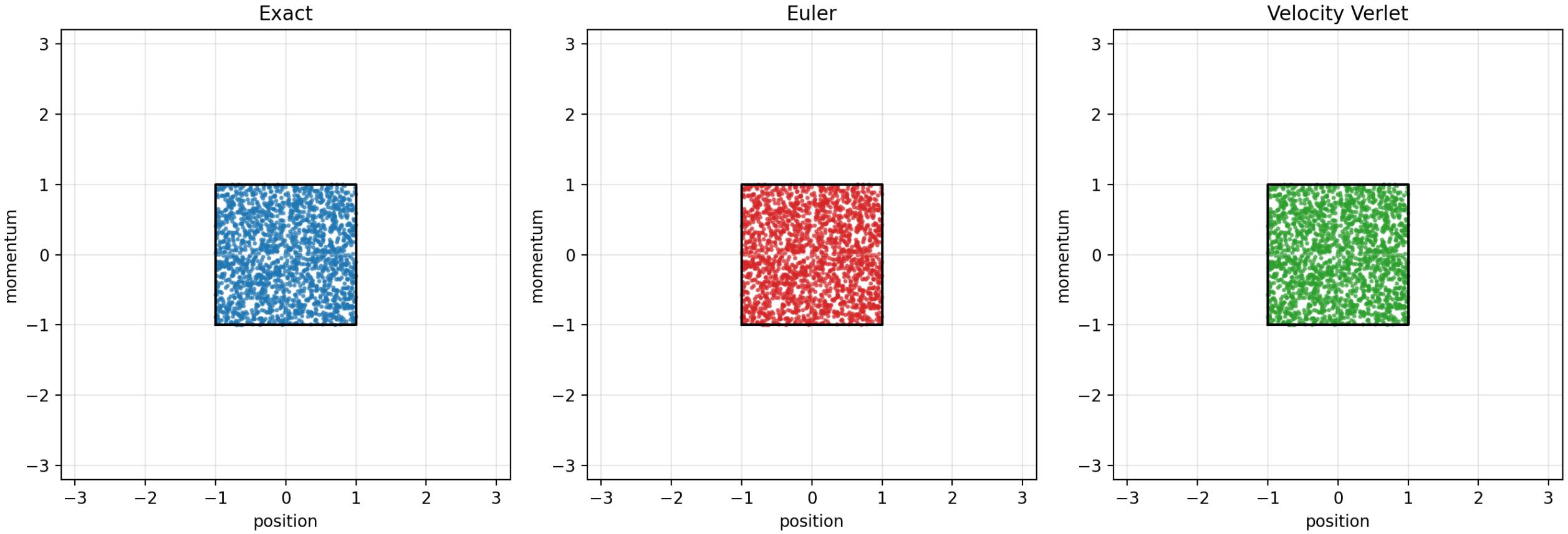


Equations of motion





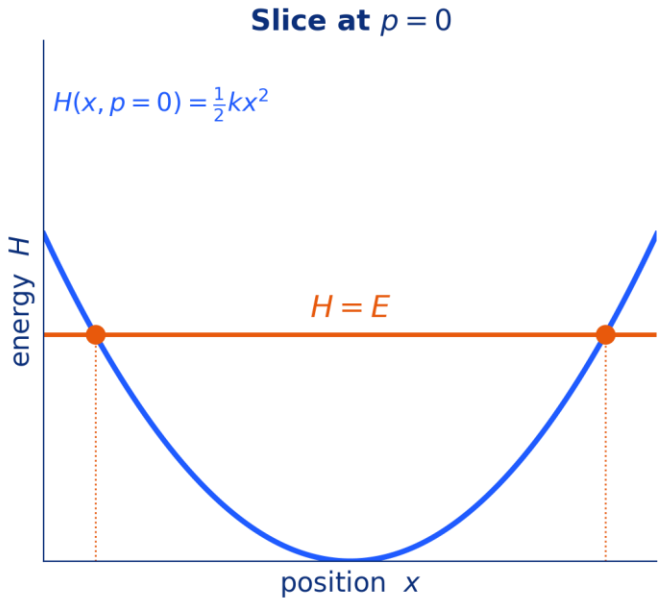
Equations of motion

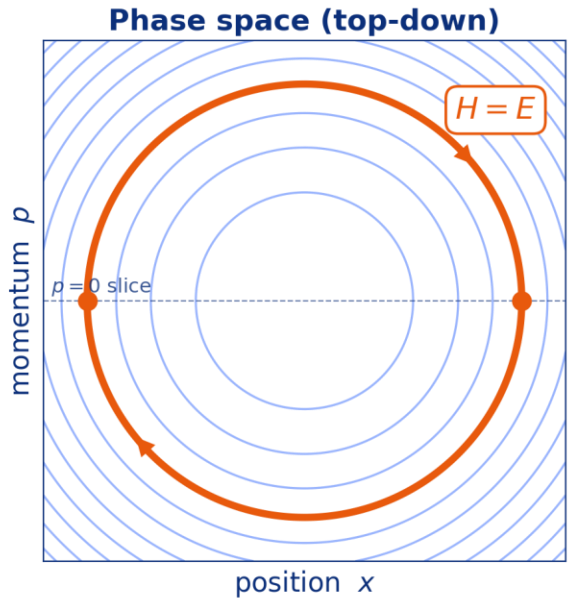


Ergodicity (NVE)

- With an integrators we have the ingredients to run an NVE MD simulation and generate a time series of atom positions and momenta.
- But how do we get the ensemble averages we are after?

NVE $$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

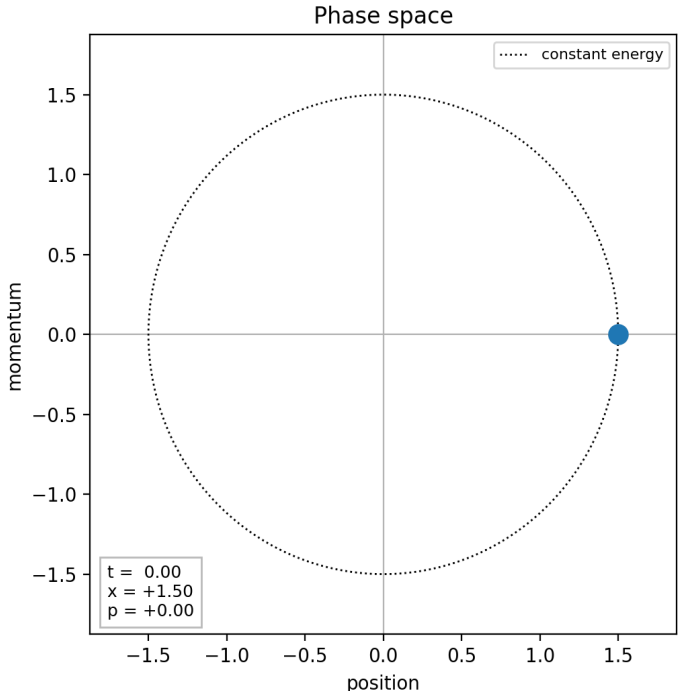


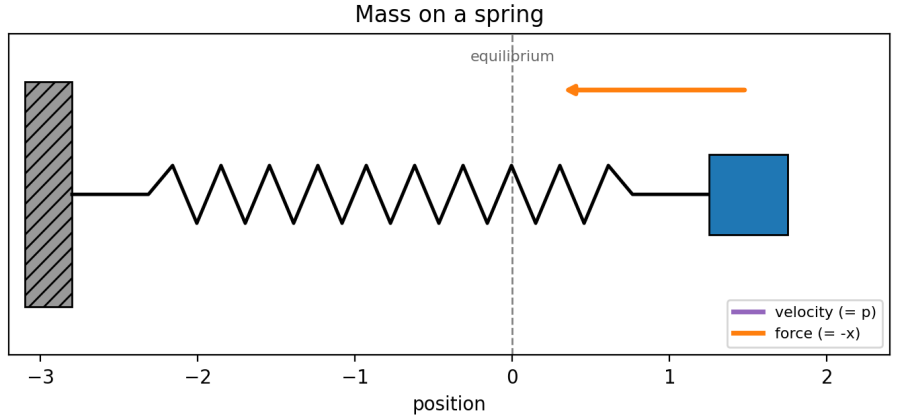


Ergodicity (NVE)

- We know from earlier when we run MD, we generate a trajectory at fixed energy.
- The points of phase space the trajectory explores will be on the constant energy phase space surface.

NVE $$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$





Ergodicity (NVE)

- If we then assume that trajectory we generate explores all parts of the fixed energy phase space, we can therefore replace our ensemble average with a time average.
- If this is true, then we say that the trajectories is ergodic and we can therefore replace our ensemble average with a time average.

NVE $$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

$$\langle a \rangle = \lim_{T\to\infty} \frac{1}{T} \int_0^T \mathrm{d}t\ a(\mathbf{x}_t)$$

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- If this is true, then we say that the trajectories is ergodic and we can therefore replace our ensemble average with a time average.
- When we run MD when generate a discretized trajectory. We therefore must approximate our time integral with a summation.

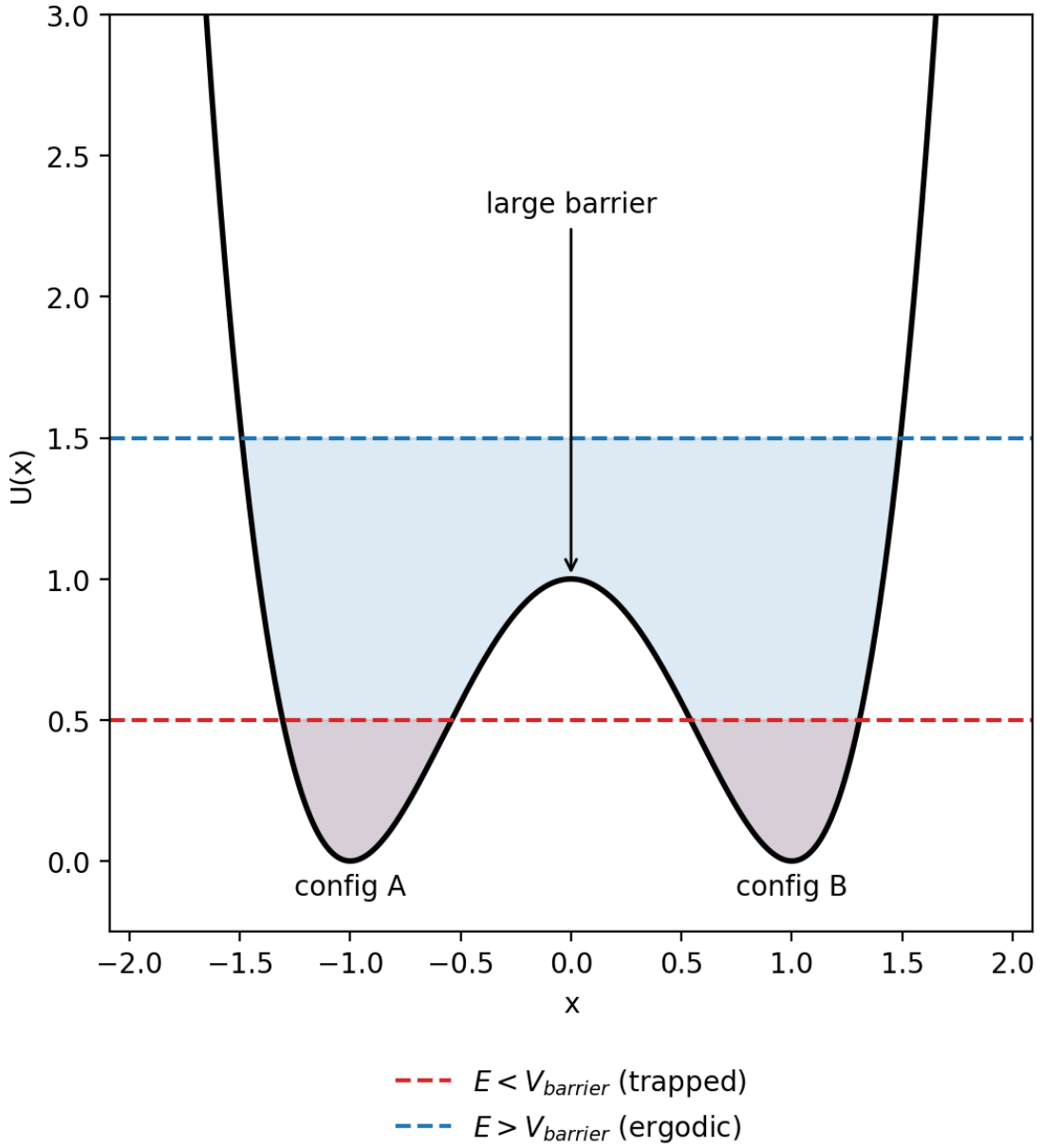
NVE $$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

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$$\langle a \rangle = \lim_{M\to\infty} \frac{1}{M} \sum_{t=0}^{M} a(\mathbf{x}_t)$$

Ergodicity (NVE)

- How do we know whether the trajectories we produce will be ergodic? In general, we don't know and there are examples where we would not expect it to be true or at least not true for the time periods that we can simulate.
- For example, two equal energy configurations separated by a very large potential energy barrier.



Ergodicity (NVE)

- From the previous slides we therefore have all the ingredients to sample the NVE ensemble by evolving the trajectory over time at constant energy.

NVE $$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})\delta(H(\mathbf{x}) - E)}{\int \mathrm{d}\mathbf{x}\ \delta(H(\mathbf{x}) - E)}$$

Symplectic integrator

$$q(t + \Delta t) = q(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2$$

$$v(t + \Delta t) = v(t) + \frac{\Delta t}{2}[a(t) + a(t + \Delta t)]$$

Ergodic trajectory

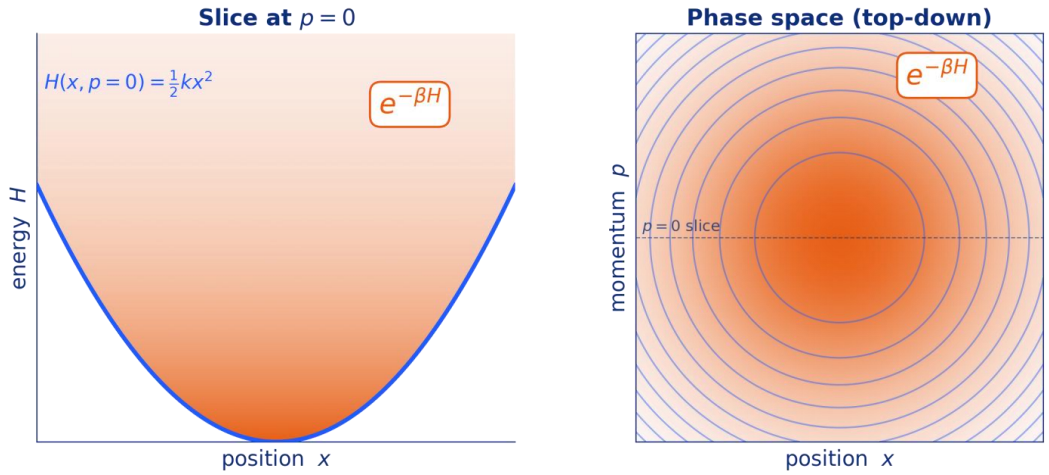
$$\langle a \rangle = \lim_{T\to\infty} \frac{1}{T} \int_0^T \mathrm{d}t\ a(\mathbf{x}_t)$$

Temperature

- However, to match some experiment or to obtain some specific property we may want to run MD at some fixed temperature.
- We therefore need some mechanism to fix/maintain the temperature of the simulation but also sample phase space following the distribution f(x).
- To sample phase space a fixed temperature means we need some scheme which generates trajectories with variable total energies.

NVT $$\langle a \rangle = \frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x}) e^{-\beta H(\mathbf{x})}}{\int \mathrm{d}\mathbf{x}\ e^{-\beta H(\mathbf{x})}}$$

$$\langle a \rangle = \lim_{M \to \infty} \frac{1}{M} \sum_{t=0}^{M} a(\mathbf{x}_t)$$



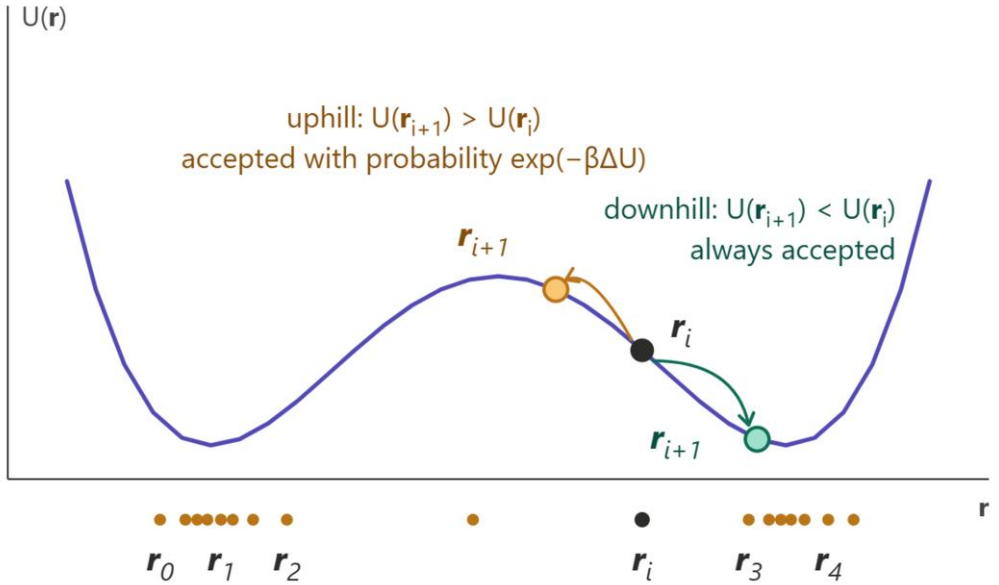
Temperature

- To sample phase-space a fixed temperature means we need some scheme which generates trajectories with variable total energies.
- Think back to the Monte-Carlo slides. The sampling of phase-space was carried so that following a specific distribution with greater sampling at lower energies. Our MD trajectories will need to do something similar.

$$A(\mathbf{r}'|\mathbf{r}) = \min\left[1, e^{-\beta[U(\mathbf{r}')-U(\mathbf{r})]}\right]$$

$$\langle b \rangle = \int \mathrm{d}\mathbf{r}\ b(\mathbf{r}) f(\mathbf{r}) = \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} b(\mathbf{r}_i)$$

$$f(\mathbf{r}) = \frac{e^{-\beta U(\mathbf{r})}}{\int \mathrm{d}\mathbf{r}\ e^{-\beta U(\mathbf{r})}}$$



Temperature

- We will start by having a look at the Nosé Hamiltonian.

$$H_{\mathrm{N}}(\mathbf{q}_1, \ldots, \mathbf{p}_1, \ldots, s, p_s) = \sum_i^N \frac{\mathbf{p}_i^2}{2m_i s^2} + U(\mathbf{q}_1, \ldots) + \frac{p_s^2}{2Q} + g k_{\mathrm{B}} T \ln(s)$$

Temperature

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First note the two additional coordinates: s(t) and its conjugate momentum p_s(t).

Temperature

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The coordinate s(t) scales the momenta.

Temperature

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$$H_{\mathrm{N}}(\mathbf{q}_1, \ldots, \mathbf{p}_1, \ldots, s, p_s) = \sum_i^N \frac{\mathbf{p}_i^2}{2m_i s^2} + U(\mathbf{q}_1, \ldots) + \frac{p_s^2}{2Q} + g k_{\mathrm{B}} T \ln(s)$$

Momentum

Potential

The coordinate s(t) follows an equations of motion with its own momenta and potential terms.

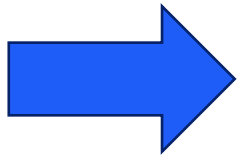
Temperature

- An NVE ensemble average with the Nosé Hamiltonian would be equivalent to NVT with the normal Hamiltonian after the extra coordinates are integrated away.

$$H_{\mathrm{N}}(\mathbf{q}_1,\ldots,\mathbf{p}_1,\ldots,s,p_s)=\sum_i^N\frac{\mathbf{p}_i^2}{2m_is^2}+U(\mathbf{q}_1,\ldots)+\frac{p_s^2}{2Q}+gk_{\mathrm{B}}T\ln(s)$$

NVE

$$\frac{\iiint \mathrm{d}\mathbf{x}\mathrm{d}s\mathrm{d}p_s\ a_{\mathrm{N}}(\mathbf{x})\delta(H_{\mathrm{N}}(\mathbf{x},s,p_s)-E)}{\iiint \mathrm{d}\mathbf{x}\mathrm{d}s\mathrm{d}p_s\ \delta(H_{\mathrm{N}}(\mathbf{x},s,p_s)-E)}$$



NVT

$$\frac{\int \mathrm{d}\mathbf{x}\ a(\mathbf{x})e^{-\beta H(\mathbf{x})}}{\int \mathrm{d}\mathbf{x}\ e^{-\beta H(\mathbf{x})}}$$

Temperature

- From the Nosé Hamiltonian we can generate a set of equations of motion. However, they're not convenient for MD simulations.

$$\frac{\mathrm{d}\mathbf{q}_i}{\mathrm{d}t} = \frac{\partial H_N}{\partial \mathbf{p}_i} = \frac{\mathbf{p}_i}{m_i s^2}$$

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \frac{\partial H_N}{\partial p_s} = \frac{p_s}{Q}$$

$$\frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = -\frac{\partial H_\mathrm{N}}{\partial \mathbf{q}_i} = -\frac{\partial U}{\partial \mathbf{q}_i}$$

$$\frac{\mathrm{d}p_s}{\mathrm{d}t} = -\frac{\partial H_\mathrm{N}}{\partial s} = \frac{1}{s}\left[\sum_i^N \frac{\mathbf{p}_i^2}{m_i s^2} - g k_\mathrm{B} T\right]$$

Temperature

- By applying a set of noncanonical change of variables we obtain can the Nosé-Hoover equations.

$$\frac{\mathrm{d}\mathbf{q}_i}{\mathrm{d}t} = \frac{\mathbf{p}_i}{m_i}$$

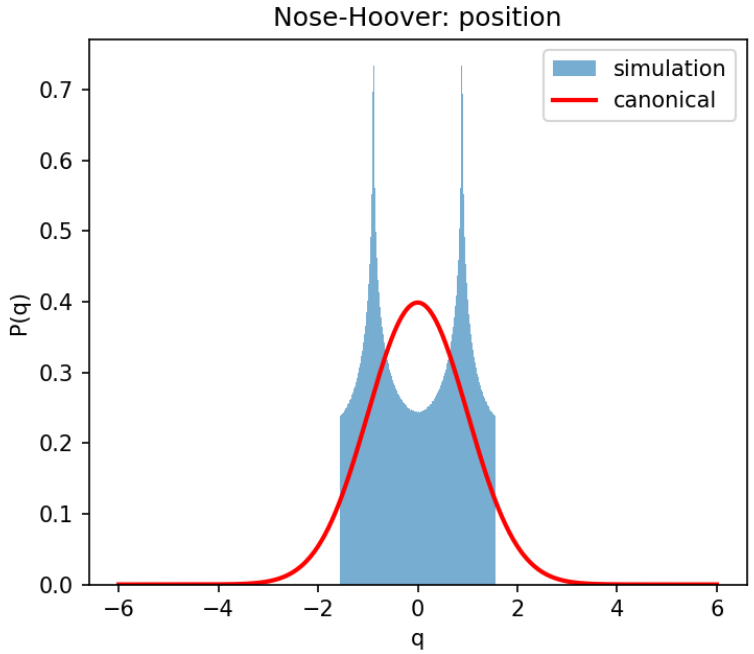
$$\frac{\mathrm{d}\eta}{\mathrm{d}t} = \frac{p_\eta}{Q}$$

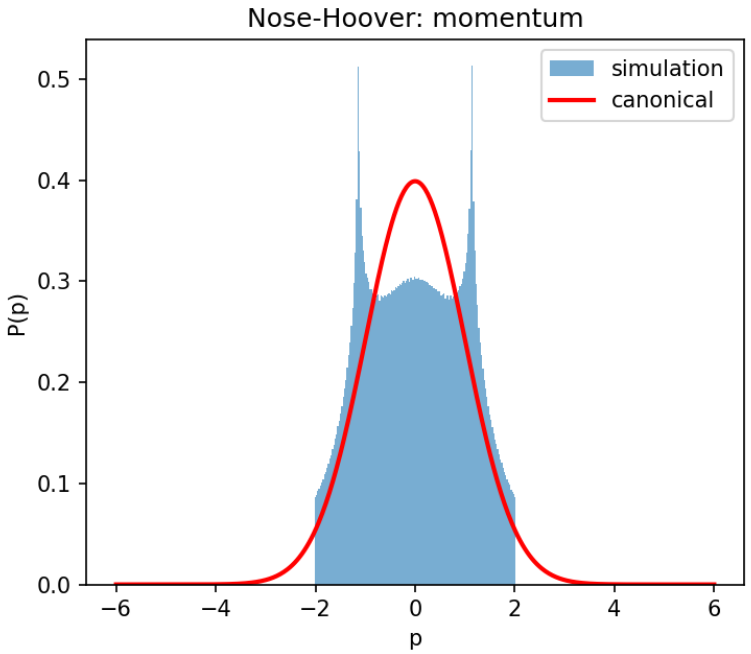
$$\frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = -\frac{\partial U}{\partial \mathbf{q}_i} - \frac{p_\eta}{Q}\mathbf{p}_i$$

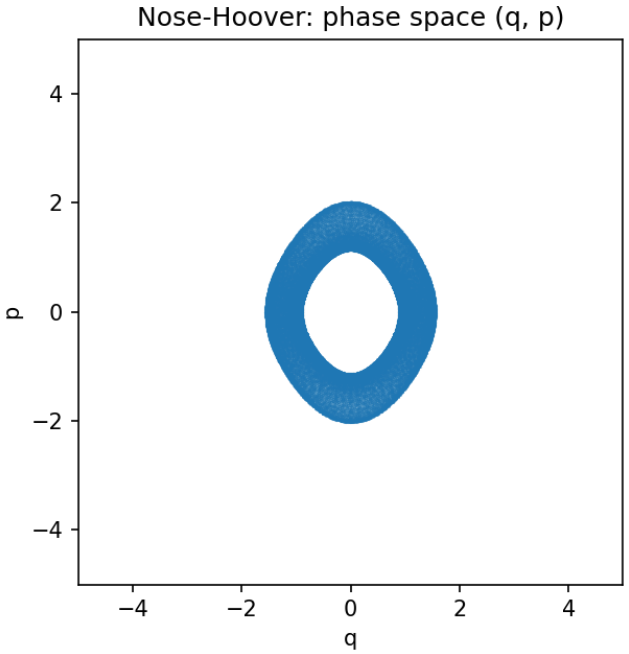
$$\frac{\mathrm{d}p_\eta}{\mathrm{d}t} = \sum_i^N \frac{\mathbf{p}_i^2}{m_i} - 3Nk_\mathrm{B}T$$

Temperature

- By applying a set of noncanonical change of variables we obtain can the Nosé-Hoover equations. Unfortunately, it does not generate the correct position and momentum distributions for some simple cases like the harmonic oscillator.







Temperature

- The Nosé-Hoover equations can be generalised by coupling the thermostats variables with additional thermostats variables in a chain.

$$\frac{\mathrm{d}\mathbf{q}_i}{\mathrm{d}t} = \frac{\mathbf{p}_i}{m_i} \qquad \frac{\mathrm{d}\eta_j}{\mathrm{d}t} = \frac{p_{\eta_j}}{Q_j}$$

$$\frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = -\frac{\partial U}{\partial \mathbf{q}_i} - \frac{p_{\eta_1}}{Q_1}\mathbf{p}_i$$

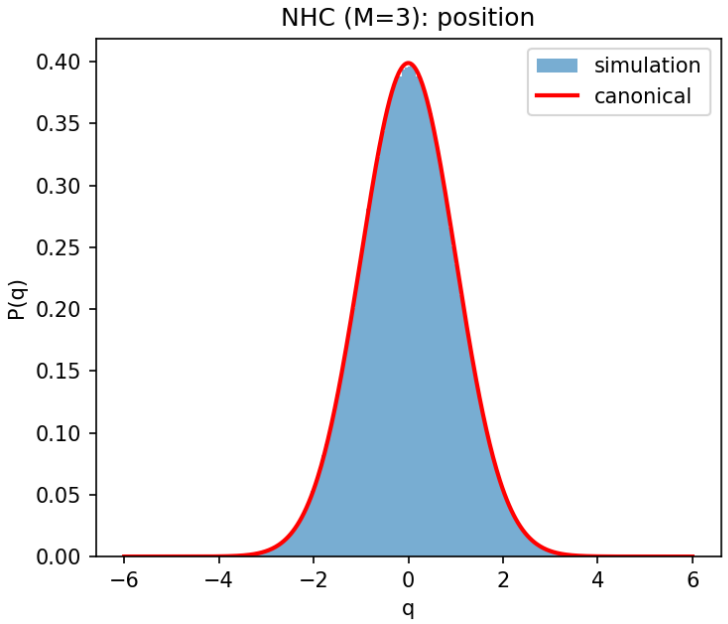
$$\frac{\mathrm{d}p_{\eta_1}}{\mathrm{d}t} = \left[\sum_i^N \frac{\mathbf{p}_i^2}{m_i} - 3Nk_\mathrm{B}T\right] - \frac{p_{\eta_2}}{Q_2}p_{\eta_1}$$

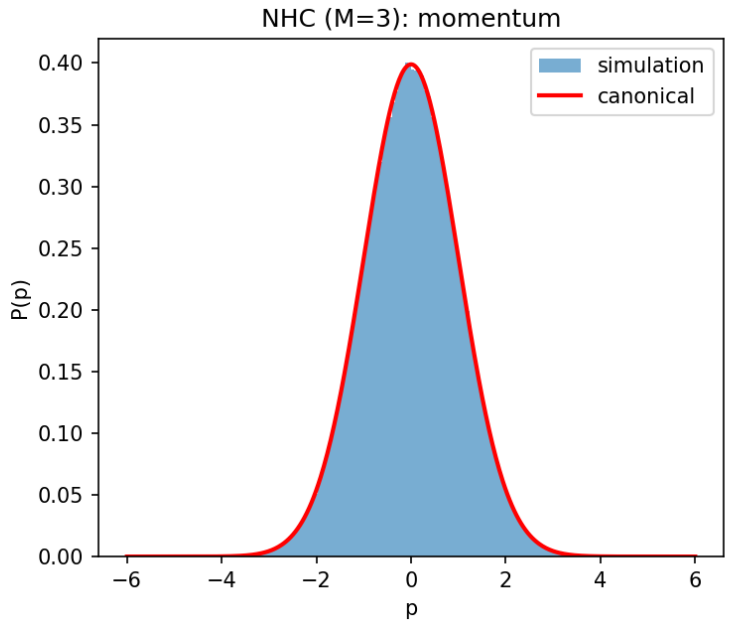
$$\frac{\mathrm{d}p_{\eta_j}}{\mathrm{d}t} = \left[\frac{p_{\eta_{j-1}}^2}{Q_{j-1}} - k_\mathrm{B}T\right] - \frac{p_{\eta_{j+1}}}{Q_{j+1}}p_{\eta_j}$$

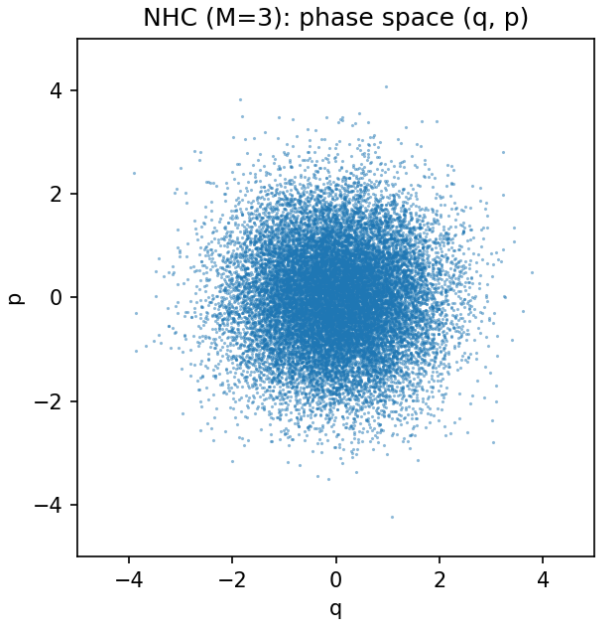
$$\frac{\mathrm{d}p_{\eta_M}}{\mathrm{d}t} = \left[\frac{p_{\eta_{M-1}}^2}{Q_{M-1}} - k_\mathrm{B}T\right]$$

Temperature

- The Nosé-Hoover equations can be generalised by coupling the thermostats variables with additional thermostats variables in a chain. This corrects the distributions for the harmonic oscillator.







Temperature

- There are a few scheme to integrate the Nosé-Hoover chain equations, for example, BAOAB. In OpenMM they use the Leap-Frog middle algorithm [1,2].

$$\mathbf{v}_i(t + \Delta t/2) = \mathbf{v}_i(t - \Delta t/2) + \mathbf{a}_i(t)\Delta t$$

$$\mathbf{r}_i(t + \Delta t/2) = \mathbf{r}_i(t) + \frac{1}{2}\mathbf{v}_i(t + \Delta t/2)\Delta t$$

$$\mathbf{v}_i(t + \Delta t/2) \rightarrow \mathbf{v}'_i(t + \Delta t/2) \qquad \text{thermostat}$$

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t + \Delta t/2) + \frac{1}{2}\mathbf{v}'_i(t + \Delta t/2)\Delta t$$



[1] https://docs.openmm.org/latest/userguide/theory/04_integrators.html
[2] Zhijun Zhang, Xinzijian Liu, Kangyu Yan, Mark E. Tuckerman, Jian Liu; Unified Efficient Thermostat Scheme for the Canonical Ensemble with Holonomic or Isokinetic Constraints via Molecular Dynamics. *J. Phys. Chem. A* 18 July 2019; 123 (28): 6056–6079.

Temperature

- There are a few scheme to integrate the Nosé-Hoover chain equations, for example, BAOAB. In OpenMM they use the Leap-Frog middle algorithm [1,2].
- The equations to integrate over the thermostat parameters so that we can scale the velocities is a bit complicated. Here we include the simplest case without the Suzuki–Yoshida scheme or multiple time step chain propagation.
- For a chain with length of 1 we get the Nosé-Hoover thermostat.

$$p_{\eta_M}(t+\Delta t/2) = p_{\eta_M}(t) + \frac{\Delta t}{2} G_M(t)$$

$$p'_{\eta_{M-1}}(t) = p_{\eta_{M-1}}(t) \exp\left[-\frac{\Delta t}{4}\frac{p_{\eta_M}(t+\Delta t/2)}{Q_M}\right]$$

$$p'_{\eta_{M-1}}(t+\Delta t/2) = p'_{\eta_{M-1}}(t+\Delta t/2) + \frac{\Delta t}{2} G_{M-1}(t)$$

$$p''_{\eta_{M-1}}(t+\Delta t/2) = p'_{\eta_{M-1}}(t+\Delta t/2) \exp\left[-\frac{\Delta t}{4}\frac{p_{\eta_M}(t+\Delta t/2)}{Q_M}\right]$$

$$\cdots$$

$$\mathbf{p}_i(t+\Delta t/2) = \mathbf{p}'_i(t+\Delta t/2) \exp\left[-\Delta t \frac{p_{\eta_1}(t+\Delta t/2)}{Q_1}\right]$$

$$p'_{\eta_1}(t+\Delta t/2) = p''_{\eta_1}(t+\Delta t/2) \exp\left[-\frac{\Delta t}{4}\frac{p_{\eta_2}(t+\Delta t/2)}{Q_2}\right]$$

$$p'_{\eta_1}(t+\Delta t) = p'_{\eta_1}(t+\Delta t/2) + \frac{\Delta t}{2} G_1(t+\Delta t/2)$$

$$p_{\eta_1}(t+\Delta t) = p'_{\eta_1}(t+\Delta t) \exp\left[-\frac{\Delta t}{4}\frac{p_{\eta_2}(t+\Delta t/2)}{Q_2}\right]$$

$$\cdots$$

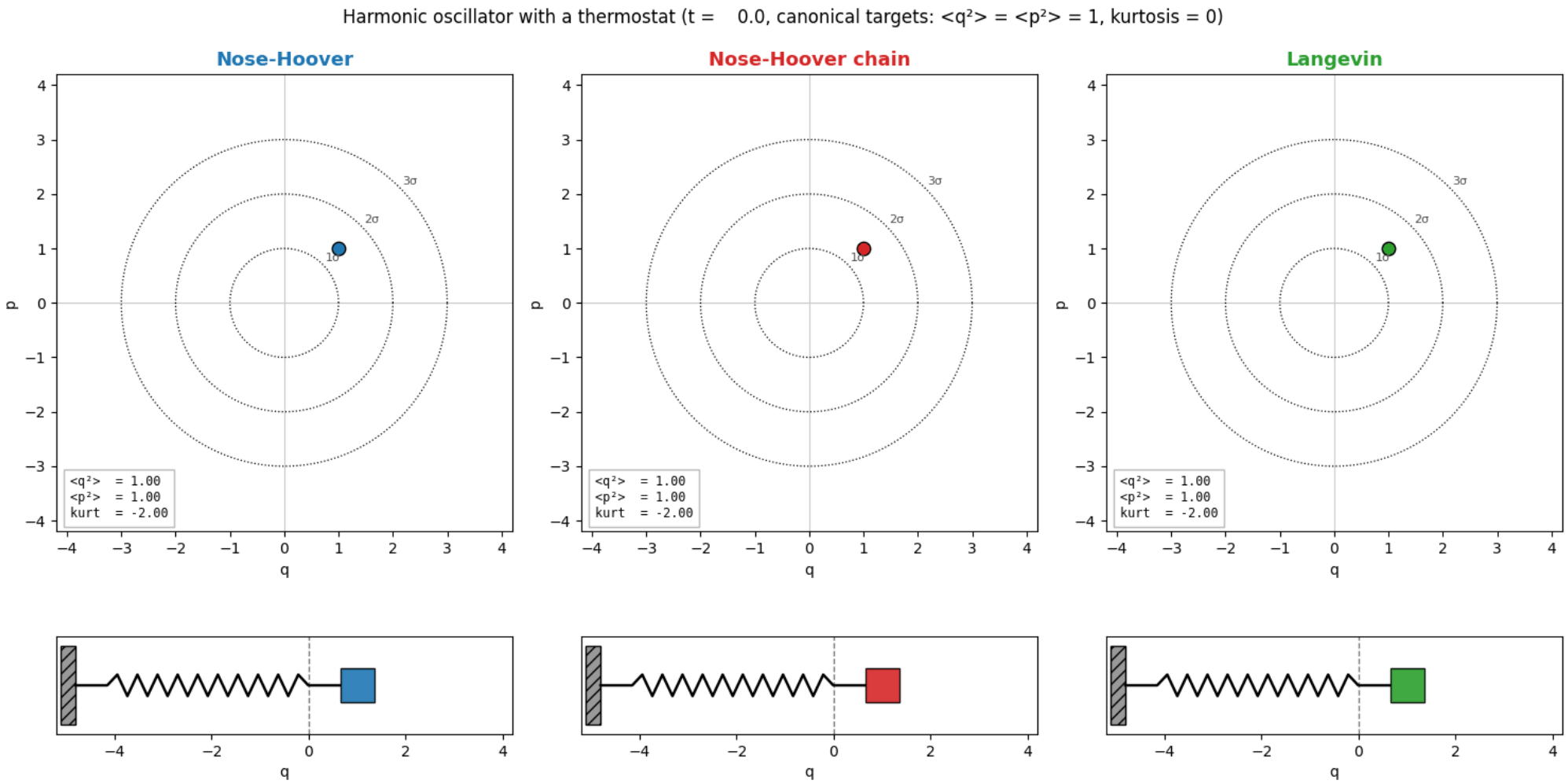
$$p_{\eta_M}(t+\Delta t) = p_{\eta_M}(t+\Delta t/2) + \frac{\Delta t}{2} G_M(t+\Delta t/2)$$

$$G_1(t) = \sum_i^N \frac{\mathbf{p}_i^2(t)}{m_i} - 3Nk_{\mathrm{B}}T$$

$$G_j(t) = \frac{p_{\eta_{j-1}}(t)}{Q_{j-1}} - k_{\mathrm{B}}T$$

[1] https://docs.openmm.org/latest/userguide/theory/04_integrators.html
[2] Zhijun Zhang, Xinzijian Liu, Kangyu Yan, Mark E. Tuckerman, Jian Liu; Unified Efficient Thermostat Scheme for the Canonical Ensemble with Holonomic or Isokinetic Constraints via Molecular Dynamics. *J. Phys. Chem. A* 18 July 2019; 123 (28): 6056–6079.

Temperature



Next steps

- Hopefully, this was a good short introduction to the theory behind MD and good start to dive deeper into the background if you want to.
- At the very least you should now understand some of the jargon that is used, for example, integrators, NVE, NVT, sampling, ergodicity, etc.
- Note that we have only just scratched the surface is there is much more to learn including, BAOAB, barostats, PIMD, RPMD, AIMD, metadynamics, correlation functions, force fields, and more.
- Of course, you can (and many people do) just run simulations without thinking about all the theory too much.

